

Soccer Match Prediction — Methodology & Accuracy Report

A market-anchored, Dixon-Coles–based system with player, cohesion, mentality and expert layers · FIFA World Cup 2026 · Report generated 2026-07-01

STATS-LED · MINDSET-ADJUSTS · MARKET-ANCHORED

1. Philosophy in one paragraph

Predict a **distribution of scorelines**, never a single fake-precise score. Anchor every forecast to the **betting market** (Kalshi / Polymarket) because market-implied probabilities are the best-calibrated public forecast that exists — but treat the market as *crowd belief, not ground truth*. Our independent analysis (Dixon-Coles + players + cohesion + mentality + a human expert) earns its keep only at **defensible divergences**: spots where we can name a structural reason the crowd is mispricing. Bet only those; pass everything else.

2. The layered model

MARKET ANCHOR

→ de-vigged Kalshi / Polymarket implied probabilities (the prior)

+ DIXON-COLES

→ team attack/defense → expected goals λ, μ

+ FORM & DECAY

→ recent matches weighted more (time decay ξ); signal vs noise

+ PLAYERS

→ disruptor availability & impact; H2H neutralisation

+ COHESION

→ collective > sum of parts (settled XI, system, chemistry)

+ MENTALITY

→ belief/resilience compresses talent gaps (the "Uruguay principle")

+ EXPERT

→ a trusted player's eye on form/fitness/tactics (strong prior)

= SCORELINE DISTRIBUTION + ADVANCE PROBABILITY + \$100 ALLOCATION

2.1 Market anchor — crowd belief, not truth

De-vig sportsbook/market prices so they sum to 1: $\text{fair}_i = \text{implied}_i / \sum \text{implied}_j$. Prediction markets are near-fair already. The market is the number *to beat*, moved off only with a stated reason. It carries biases we exploit: **favorite-longshot**, **recency/narrative**, **sentiment-inflated popular teams** (Brazil, Argentina, England), **thin-market noise**, and it is **slow to price structural factors** (rotation, key injuries) until the team sheet drops.

2.2 Dixon-Coles statistical core

Each match = two Poisson processes. Expected goals:

$$\lambda \text{ (home)} = \alpha_{\text{home}} \cdot \beta_{\text{away}} \cdot \gamma \qquad \mu \text{ (away)} = \alpha_{\text{away}} \cdot \beta_{\text{home}}$$
$$\alpha = \text{attack rating} \cdot \beta = \text{defense weakness} \cdot \gamma = \text{home advantage}$$

Two corrections over naïve Poisson:

- **τ low-score correction (ρ):** reweights 0-0, 1-0, 0-1, 1-1 — real football has more cagey low-scoring games than independence implies. We use $\rho \approx -0.06$.
- **Time decay (ξ):** each past match weighted $\exp(-\xi \cdot \Delta t)$ — recent form matters more.

2.3 Advance probability (knockouts)

Markets quote *to-advance* (incl. ET/penalties), not 90-minute results. We convert:

$$\text{advance_home} = P(\text{win } 90') + P(\text{draw}) \cdot q \quad q = 0.5 + 0.5 \cdot (\text{winFav} - \text{winDog}) / (\text{winFav} + \text{winDog})$$

The better side wins the shootout/ET slightly more often. This fix alone closed most of the early gap between our numbers and the market on favorites.

2.4 Player / disruptor layer

Rate each likely starter's **disruption degree (1–10)** — capacity to swing a tight game (Messi/Mbappé 10; Kane/Bellingham/Musiala/Yamal/Haaland 8–9; Pulisic/Cherki/James "relatively" 6–7). Then: (1) **availability gate** — not playing $\rightarrow 0$; (2) **form multiplier** — last-10 output, opponent-adjusted; (3) **H2H neutralisation** — net disruption that survives the matchup drives the edge, converted to a $\pm\lambda$ adjustment. In knockouts, individual match-winners weigh more than in league play.

2.5 Cohesion & 2.6 Mentality (the Uruguay principle)

Cohesion = how well a team plays as a unit (settled XI, same-club spine, drilled system) — mainly lowers goals conceded and lifts draw probability (Morocco, Iran, Japan, Cape Verde). **Mentality** = belief/resilience under pressure; it compresses talent gaps and raises draw/penalty likelihood. Both mean: *do not over-price a favorite against a high-belief, organised underdog; widen the distribution*. This is a source of irreducible uncertainty — expressed as a flatter distribution, not false precision.

2.7 Expert-in-the-loop & 2.8 Goals by half

A trusted human read (a current/former player) is a first-class input on form, fitness, tactics and mentality — a strong Bayesian prior that can override lagging stats. Goals split $\approx$ **45% first half / 55% second half**; early-goal probability $= 1 - e^{-(\lambda_h + \lambda_a) \cdot 15/90}$.

2.9 Allocation & staking (\$100 per match)

Edge is measured **vs the de-vigged market**: if our probability equals the market's, edge = 0 and the correct stake is 0 (agreeing with the price just pays the vig). For each candidate bet at price m (odds $o = 1/m$), our prob p :

$$\text{edge} = p - m \quad \text{EV}(\$1) = p \cdot (o-1) - (1-p) \quad [>0 \Rightarrow \text{"bargain"}] \quad \text{Kelly } f = \text{edge} / (o-1) \\ (\text{use } \tfrac{1}{4} \cdot f)$$

We present **four allocation styles** per game — Max-EV, Max probability-of-profit, Balanced, Low-scoring — because *maximising EV and maximising P(profit) rarely point at the same bet*. Caps: $\leq 5\%$ bankroll/day,

$\leq 2.5\%$ /game. The durable edge lives in the **to-advance markets on over-priced favorites**, not in exact scores or totals (which are ~fairly priced).

3. Workflow — how a match is analysed

1. Pull the live Kalshi/Polymarket board (to-advance % + exact scores + totals). De-vig.
2. Build λ , μ from strength/form + reconcile to the market's totals/team-totals.
3. Adjust for players (availability, disruptors, H2H), cohesion, mentality; fold in the expert read.
4. Compute the scoreline grid → top scenarios, W/D/W, advance %, O2.5, BTTS, goals by half.
5. Compare to the market; deviate only with a named structural reason.
6. Build the four \$100 allocations; recommend one by risk profile; log the pick before kickoff.
7. After the game: grade result, exact score, Brier, and bet P&L; feed lessons back.

4. Prediction accuracy history

4.1 June 27 — final group matchday (6 games)

First live test. Three games market-anchored, three model-only.

Match	Predicted	Actual	Result	Note
Panama–England	0–3	0–2	✓	ENG win; margin off
Croatia–Ghana	1–0	2–1	✓	CRO win; Under 2.5 lost (3 g)
Colombia–Portugal	1–2	0–0	✗	Draw; Over/BTTS lost
DR Congo–Uzbekistan	1–1	3–1	✗	Passed (0 stake)
Algeria–Austria	1–2	3–3	✗	Passed (0 stake)
Jordan–Argentina	0–2	1–3	✓	ARG win; rotation trigger stood down

Grade: results 3/6, exact scores 0/6, mean Brier $\approx$ 0.46. Staked P&L **–2.0 units** (both active bets lost).

Lesson: the Colombia–Portugal 0-0 was a cagey dead-rubber I flagged but didn't back — the contrarian read was right and I didn't trust it. Down-weight Over/BTTS in no-stakes seeding games.

4.2 Round of 32 — first half (7 games, Jun 28–30)

Game	Result	Advanced	Our bet	Net (if \$100)
South Africa–Canada	0–1	Canada	pass	—
Brazil–Japan	2–1	Brazil	pass (no edge)	—
Germany–Paraguay	1–1 → Paraguay	Paraguay	Paraguay-adv @15%	+\$260
Netherlands–Morocco	1–1 → Morocco (pens)	Morocco	Morocco-adv @38%	+\$204
Côte d'Ivoire–Norway	1–2	Norway	marginal pass	—
France–Sweden	3–0	France	alloc C (hedged)	+\$9
Mexico–Ecuador	2–1	Mexico	Ecuador-adv (alloc D)	–\$100
Slate	\$400 staked			+\$373 (+93%)

Headline: both "fade the over-priced favorite" bets hit — **Morocco** (crowd 38% → we said 47%; won on penalties, Bono the hero) and **Paraguay** (crowd 15% → we said 24%; Germany knocked out). The no-edge games were correctly passed; the diversified France allocation banked a small win off the exact-score hedge even though the Sweden thesis was wrong.

4.3 Honest calibration notes

- **Process worked:** the system located the two mispriced favorites, bet the +EV underdog, and passed the efficient lines. Betting edges 3 of 4 profitable.
- **Respect variance:** Morocco won on penalties and Paraguay went to ET — near coin-flips in the moment; we caught the good side. The bets were +EV (the durable part); a +93% slate regresses over time.
- **The loss is instructive:** Mexico–Ecuador was the *lower-conviction* fade (Mexico's altitude/home edge was real) — flagged "don't oversize," and it lost. Conviction ranking matters.
- **Where the edge lives:** to-advance markets on over-priced favorites — not scorelines or totals, which the market prices efficiently (often richer than our model).

4.4 Running scorecard

Slate	Result hit	Exact score	Staked P&L	Verdict
Jun 27 group (6)	3/6	0/6	-2.0 u	market beat us on anchored games
R32 first half (7)	5/7 modal · 2/2 headline bets	Brazil 2-1 ✓	+\$373	both fades hit; passes correct

5. Key lessons codified

- Anchor to the market; deviate only with a nameable structural reason (crowd belief ≠ truth).
- The recurring edge is **fading sentiment-inflated favorites in to-advance markets** (validated by Morocco & Paraguay).
- Cohesive/high-mentality underdogs drag favorites to penalties — where shootout keepers (Bono) swing ties; price that in.
- Max-EV ≠ max-P(profit): always present the allocation menu, size by conviction, cap exposure.
- Dead-rubber / seeding games skew low-scoring — down-weight Over/BTTS.
- Grade everything (Brier + P&L) before the next slate; that feedback is the system.

6. Repository map

All analysis is versioned on branch `claude/soccer-prediction-research-payza0`.

File	Contents
01-methodology.md	Full theory: Dixon-Coles, market layer, players, cohesion, mentality, expert
02-prediction-prompt.md	Portable, reusable prediction prompt
05 / 13	Disruptor rubric + worked example; match-day analysis playbook
07-predict-and-score-log.md	Every prediction graded (Brier + P&L)
12 / 17	R32 scoreline scenarios; results report + realigned predictions
15 / 16	Match-day allocations & final reports (Jun 29–30 slate)
tools/*.py	Poisson/DC engine, allocation evaluator, odds connector